1.4b The Brain

Brain Regions and Structures

What are the main sections in the Brain?

Vertebrate brains have 3 main sections

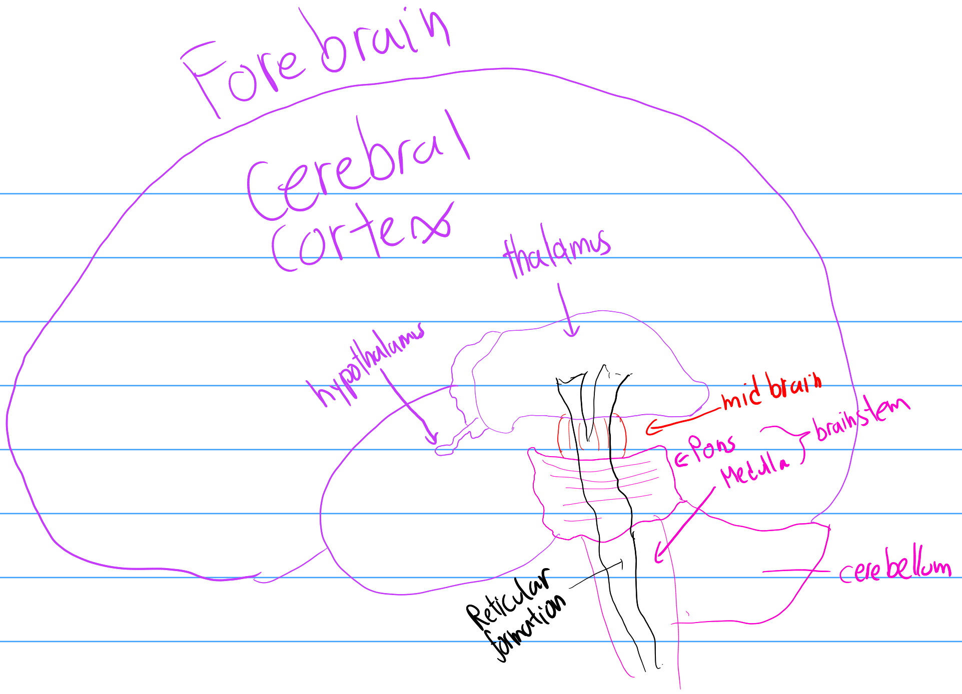
The Hind brain contains brainstem structures that direct essential survival functions like breathing, sleeping, arousal, coordination, balance, etc.

The midbrain is atop the brainstem. Connects Hbrain with Fbrain. Controls some movement and transmits info enabling seeing & hearing.

The fore brain manages complex cognitive tasks, sensory & associative functions, and voluntary motor activities.

Different organisms will vary in development in these areas

Humans have very developed forebrains, for complex desicions
Some Sharks have complex hindbrains, so they can chase prey



What is the Brainstem?

The brainstem is the brain's innermost region. responsible for automatic survival functions.

Its base is the medulla, the slight swelling in the spinal cord just after it enters the skull. The medulla controls heartbeat & breathing.

Pons, ↑ medulla, helps coordinate movement & control sleep

With just a brainstem, in theory one could survive and move, but could not do so purposefully.

The brainstem is a crossover point. Most nerves to & from each side of brain connect with the body's opposite side here because of the brain's contralateral hemispheric organization.

Near the brainstem

The Thalamus, ↑ brainstem, is a pair of egg-shaped structures that act as the brain's sensory control center.
It recieves info ∀ senses except smell.
routes that info to respective regions
It also receives replies from those regions which it routes to the medulla & cerebellum.

The reticular formation ('netlike') is inside the brainstem.
Nerve network governed by the Reticular Activating System
Extends from spinal cord upto thalamus.
Some spinal cord sensory input flows through RF, which filters incoming stimuli, and relays important info to other brain areas.
Controls arousal, our state of alertness.

Cerebellum → brainstem. Baseball-sized.
Two wrinkled halves resemble "little brain". It, (+ basal ganglia) enables nonverbal and skill learning. Help from Pons ⇒ Coordinates voluntary movement.
$>\frac{1}{2}$ neurons are in it.
Operates just outside your awareness.

The previous parts all focused primarily on function without concious effort.

Our brain processes most info outside our awareness.

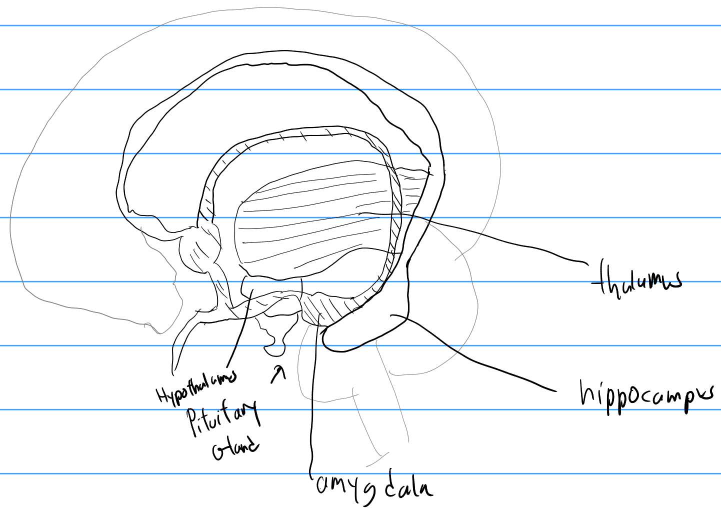
What is the Limbic System?

The Limbic system deals with emotions, motivations & memory formation.

Fear & Aggression

The amygdala, 2 lima-bean sized neural clusters, enables aggression & fear. People with amygdala lesions have sometimes not displayed fear at all.

But, amygdala is engaged with other mental phenomena. Feelings of fear & aggression cuses neural activity in other parts of the brain



The hypothalamus ⓥ thalamus. It is a link in the command chain governing bodily maintenance.

Some neural clusters in the hypothalamus influence: hunger, thirst, body temp, sexual behavior. These clusters together help maintain a homeostatic state.

It tunes into blood chem and incoming brain orders to monitor body state.

Ex.
It gets signal from CC that one is thinking of sex
∴ It secretes hormones
∴ hormones trigger PG, which controls ES to release their hormones
∴ ES hormones intensify sexual thoughts

It was found that stimulating the hypothalamus in rats was so rewarding they would self-stimulate these regions > 1000 $\frac{\text{times}}{\text{hour}}$

Some evidence indicates that humans have limbic centers for pleasure. However Stimulating our reward center areas produces more desire than enjoyment.

Memories

Hippocampus: Curved brain structure; processes conscious, explicit memories.

Lose Hippocampus (from injury, surgery, etc.) ⇒ lose ability to form memories.

Age ↑ ⇒ HC size, function ↓ ⇒ cognitive decline ↑

What is the Cerebral Cortex?

The Cerebrum: two cerebral hemispheres.
~85% of mass of brain
enables perceiving, thinking, speaking

The Cerebral Cortex covers the Cb.
A thin layer of interconnected neurons.

Humans are distinct because of the size & interconnectivity of our CC.

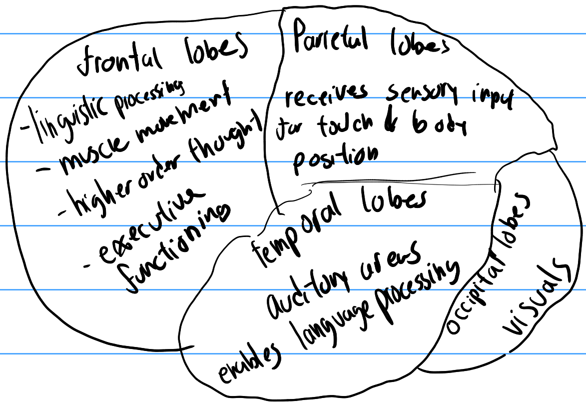
The brain looks like a wrinkled organ, shaped like an oversized walnut.

The left & right areas are primarily filled with axons connecting the cortex to other brain regions

CC contains ~20-23 billion neurons & 300 trillion synaptic connections.

Each hemisphere's cortex is subdivided into 4 lobes.

Frontal lobes -
Parietal lobes -
occipital lobes -
temporal lobes -



motor cortex at → FL.

Scientists can predict a monkey's arm motion just before it moves by measuring motor cortex activity.

Sends messages to body from MC

Trials of Cognitive Neural Prosthetics involve reading brain activity to allow people to move prosthetics with their mind.

Personal Connection

Many recent work flow changes I've made can be thought of as a primitive version of CNP. These changes, from switching my keyboard layout from Qwerty to Colemak, extensively using keyboard shortcuts, and my current research into split keyboards are all in an effort to maximize the speed of information transfer from my brain to a computer: which this tech does instantly.

Somatosensory Cortex

is the part where the CC receives messages.

← of PL, || → MC

Specifically, info from skin senses.

Sensitivity of region $\propto$ % of somatosensory devotion

CC also receives input from visual cortex in OL

sound input is processed by auditory cortex in TL

Instead of specific functions like sight, etc. association areas deal with higher mental functions like thinking or learning etc.

They cant be neatly mapped
they are found in all 4 lobes

The prefrontal cortex in the ⇐ part of FC enables judgement, planning social interactions & processing new memories.

FC helps steer us to kindness & away from violence.

PL enables mathematical & spatial reasoning.

Stimulation of a PL area caused patients to feel they had moved, but in a different area, patients moved but were unaware of it.
This suggests our perception of motion comes from intention.

↓ TL_R an association area enables face-recognition

Most complex tasks don't reside in one single area

MCQ: 1.c 2.b 3.b 4.b 5.c 6.a. 7.b. 8.d